

COMPUTER SYSTEM TROUBLESHOOTING

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COM 214

Course outline

- Understanding the process of Computer system fault diagnosis.
- Understanding computer system peripherals failures.
- Understanding Virus protection utility failure and software diagnostic tools.
- Understanding networks failure symptoms.

INTRODUCTION

Computer system comprises of several electronic components and sub-circuits that drives or control various tasks as may be required by the user. And, because these components and sub-assemblies had to work together to produce an expected outcome, it becomes very difficult sometimes to detect failure that emanates from a particular source. Even though modern components and sub-assemblies mostly come in form of field replaceable unit (FRU), wherein a user might be able to replace a component in the computer system without necessarily resorting to any professional help, the very act or process of identifying and locating the root cause of a particular failure in the computer system need some proficiency in the knowledge of the inter-relation between the hardware and software aspect of the system as related to specific functions of the computer operation.

Troubleshooting basically entails identifying symptoms of failure, detecting the cause or source of the failure, repair or replacing the faulty component or device that causes the failure and finally testing to ascertain that the problem no longer exist.

In this course, students will be exposed to various approach to solving problems arising in the operation of computer systems (hardware / software), and their peripherals. In addition they will be acquainted with the use of fault report forms that keeps record of activities done in the computer system as these records come in handy when system over hauling, or specific upgrade is required.

1.1 Various components of Computer system

There are several electronics devices or sub units (hardware) that make up the entire computer system. However, the most important and indispensable amongst them are: Motherboard (MoBo), CPU, GPU, RAM, chassis, storage device, keyboard and power supply unit. Each of the

components works together with other components and architecture to achieve specific objective demand of the computer system user.

1.2 Power On Self-Test (POST)

This referred to a process performed by a firmware or software routines immediately after a PC is powered on. The process is carried out to ensure that all the hardware is ok before the operating system is loaded. The POST operation normally runs at the beginning of the boot sequence. If there is a problem with the boot sequence, the BIOS generates a POST error code or message which is displayed on the monitor (if the VDU is already initialized), or through beeps codes. POST is the first step in hardware boot sequence therefore the PC won't continue with the boot sequence if the POST fails. If the POST passes, the PC generates a single beep sound satisfying that all is well with the PC.

1.3 Power Supply fault diagnosis

The power supply unit is the life-line of the PC. If there is a problem with this unit, the common outcome is that the PC will not turn on. Other symptoms include: the PC rebooting itself, PC turning off unexpectedly, appearance of smoke with smell of burning electronics etc.

Table 1.0 presents probable causes of PSU failure and how they can be resolved.

Possible causes of failure	Possible solution
i. The computer is not plugged into the AC outlet. ii. AC outlet is faulty. iii. Power cord is faulty. iv. Power supply switch is not turned on. v. Power supply switch is set to incorrect voltage. vi. The power button is not connected correctly to the front panel connector. vii. The power supply has failed	i. Plug the computer into a known good AC outlet. ii. Same as (i). iii. Use a known good power cord. iv. Turn on the power switch. v. set the power supply switch to the correct voltage setting. vi. Correctly orient the power button to the front case panel connector and reconnect. vii. install a known good power supply

The symptoms and possible solution presented in table 1 is typically for a situation where the computer could not turn on completely. For the other symptoms (unexpected rebooting, smoke or burning smell) where the power supply is on, it indicates that the power supply is beginning to fail and the possible remedy is to replace it out rightly.

1.4 Software diagnostic test for Hardware

In most operating systems, some of the commonly used diagnostic tools are:-

- 2 Disk error-checking or Scandisk
- 3 Disk clean up
- 4 Disk defragmenter
- 5 AIDA64Extreme (Win 10)
- 6 Crystal Disk Info
- 7 Intel Processor Diagnostic tool
- 8 JScreenFix
- 9 Benchmarking, etc.

The first three have been discussed in com 216 lecture notes.

AIDA64Extreme is a diagnostic tool associated to Windows 10. It provides detailed information about installed software and hardware (over 185k); and also offers diagnostic functions. It also offers a couple of benchmarks for measuring either the performance of individual hardware components or the whole system. It is compatible with all 32-bit and 64-bit Windows editions, including windows 11 and Windows Server2022. Other features include: hardware detection capabilities (to provide detail information on the computer internals without the need to open it up), external display support (over 50 LCD/FPD screens) and wireless remote monitoring of sensors values on smartphones and tablets, support over 250 various sensor devices to measure temperatures, voltages, fan speeds and power draw, provides detail information on installed programs, software licences, security applications and Windows settings etc.

The Crystal Disk info is good for diagnosis of sudden occurrence of fault in the PC. It performs a complete analysis of the state of your data drives including USB drives, SSDs and HDDs. It also calculates the overall health status and gives details about error rates, uptime, spin up time and temperature of the disk.

The Intel processor Diagnostic tool is best suited for the diagnosis of Intel central processing Units (CPUs).

JScreenFix tool is best for identifying LCD problems.

The benchmarking tools compare newer system to older ones with a view to establishing the overall performance of a new system before making a purchase decision. Example of such tools is: BatteryMark, Netbench, Serverbench, WebBench, WinBench97 etc.

1.5 Start up and Start up Failure

Start-up failure refers to those problems that occur during the initialization process (Boot process) of the computer; that is, between the time that the PC is powered, or turn on, up till the time that the PC sits idle at the command-line prompt or graphical desktop.

During the boot process, if all output voltages from the power supply are valid, the supply generates a power good (PG) logical signal within 100ms and 500ms. When the motherboard timer IC receives the PG signal, the timer stops forcing a reset signal to the CPU, prompting the CPU to start processing. The initial search of the address FFFF: 0000H by the CPU and the subsequent redirection of the CPU (depending on the BIOSROM) is traditionally referred to as the bootstrap.

During the initialization also, the core tests is carried out to ensure system integrity. A set of hardware specific self-test routines checks the major motherboard (MB) components and identifies the presence of any other specialized BIOS ICs in the system such as drive controller BIOS, video BIOS, SCSI BIOS, etc. System BIOS starts with a test of the MB hardware like the CPU, math-co processor, timer IC, Direct memory access (DMA) controllers and interrupt (IRQ) controllers.

The summary of the start-up process sequence is as follows:

- a) Power ON
- b) POST is conducted within 0.1 to 0.5 seconds
- c) Microprocessor timer chip receives PG signal which prompt it to stop generating reset signal to the microprocessor (CPU)
- d) Microprocessor begins executing the ROMBIOS code from memory location FFFF: 0000H
- e) ROMBIOS performs a test of the central hardware to verify basic system functionality and report any error by generating beep codes.
- f) BIOS performs video ROM scan of memory locations C000:0000 to C780:0000, looking for video adapter. If a video ROMBIOS is found, it is tested by checksum procedure. If the test is passed, the ROM is executed and the video ROM code initializes the video adapter to produce a cursor onscreen. If the checksum procedure fails, error such as "C000 ROM" or "video ROM error" may be displayed, and the system initialization usually halts.

1.5.1 Cold and Warm Start (POST)

A cold start is a situation where power is applied for the first time on a PC, whereas a warm start is by using reset button (<ctrl>+<Alt>+key combination). The memory location 0000:0472H contains a flag that determines whether initialization is a cold start or a warm start. A value of 1234h at this address indicates a warm start, in which case the POST routine is skipped. Any other value at this address is assumed to be a cold start, and full POST routine will

be executed. The full POST checks many of the higher-level functions on the MB, memory, and other sub-systems. Dozens of test are performed by the full POST.

If the POST completes successfully, the system will respond with a single beep from the speaker.

1.5.2 Causes of start-up Failure

A number of reasons could be responsible for start-up failure. Some of these reasons include:

- Power supply problems
- Inability of the system to carry out POST successfully
- Failure to load an operating system (OS) due to issues related to IO.SYS, MSDOS.SYS or DOS volume boot sector (VBS); normally accompanied with errors like "Diskette boot record error" or "Non-system disk or disk error" for example when the boot sector is corrupted this could happen.
- When 'Devices referenced in SYSTEM.INI or Registry could not be found' Note that the most common causes of this referencing error is deleting an application rather than uninstalling it. Uninstalling an application removes the files from the computer, its references are removed from the registry, and SYSTEM.INI files and its listing in the programs menu is removed. Simply deleting an application would keep these references in the system. Always remove applications using manufacturers uninstall utility or the Windows uninstall utility, located in the Add/remove programs dialog box.
- Corruption of Swap files - "swap file corrupt" leading to frequently 'hanging' or locks up (A swap file is a space created in a hard disk to store temporary data that will not fit in regular RAM. However, this space can also be corrupted and present start-up problems). In some Windows versions like 9x and 2000, the remedy could just be to simply delete the swap file.

Remedies

In most Windows versions, the following troubleshooting guide could help to resolve these file problems:

- Open SYSTEM.INI
- Determine where the referenced file should be located
- Check for the presence of that file in the proper location
- If the file does not exist, reinstall or copy it; it is possible it might have been corrupted and will have to be replaced.
- Restart the computer.
- However, if the problem persist or if the referenced file is not needed, Open the SYSTEM.INI file

➤ Place a semicolon (;) in front of the line referencing the problem file. This symbol will instruct Windows to skip over the line during start up.

1.6 The cause of Hard disk drive Failure

1.6.1 Introduction

The hard disk drive (HDD) is a non-volatile storage device that stores digitally encode data. In disk storage mechanism, data is recorded either through electronic, magnetic, optical or mechanical changes to a surface layer of one or more rotating disks. A disk drive is a device that implements such storage mechanism. Example of disk drives includes:

Floppy disk drives

Hard disk drives (non-removable)

Optical disk drives, and

Associated disk media.

1.6.2 Types of HDD

HDD are usually identified by their kind of connectors or interface. They are:

i. IDE / PATA (integrated device electronics / parallel advance technology attachment):-

This has 40 pin ribbon connectors and can connect up to 2 devices.

ii. SATA: - This has 7 conductors that connect a single HDD to a single connector on the

SATA controller found on the MB.

iii. SCSI: - This has 50 – 68 pin, multiple SCSI devices can be connected to a single channel

through daisy chaining. There is also the serial attached SCSI, or SAS for short, which is

a point – to – point serial protocol that moves data to and from computer-storage devices

such as hard disk; which has replaced the older parallel SCSI bus technology that first

appeared in the mid-1980s.

iv. External USB: - This is essentially a PATA or SATA device in a simple enclosure with a

conversion for connection to USB interface.

1.6.3 Connection of HDDs

Two HDDs may be connected to a single IDE connector using jumpers, in the following mode:

primary master – secondary master, or primary slave – secondary slave.

A single HDD however, can be connected either as primary master slave or secondary master slave (using jumper settings and choosing IDE connection).

With the advent of SATA (Serial ATA) Technology, which is now the standard for connecting hard drives and SSDs, the master/slave has been eradicated. SATA drives are automatically recognised and assigned their roles by the computer BIOS.

1.6.4 Reasons for HDD failure

HDD failure is basically categorized under four (4) broad sections viz- a- viz firmware corruption e.g driver issues, electronic failure e.g spike on controller board, mechanical failure e.g head crash and logical failure e.g invalid entry in FAT. Other reasons include:

1. Poor power supply
2. Virus attack
3. Corrupted OS
4. Damaged blocks
5. Air filters problems etc.

1.6.5 Common symptoms of HDD failure and Remedy

SYMPTOMS	POSSIBLE CAUSE	REMEDY
1. Corrupted files on HDD	Unethical shutting down, closing a running program accidentally, power failure or corrupted software use	Follow the right procedure for shutting down, avoid installing suspected software and malicious programs, clean and remove unwanted programs regularly
2. Virus manifestation	Virus could attack and spread from HDD to other systems across the network of computers.	Update your system regularly with the help of reliable anti-virus programs.
3. Unexpected PC crashes	HDD has too many bad sectors or blocks such that the read head may not be able to identify the sector requested by the user, or the head makes contact with the platter, producing clicking sounds. Old age, formatting HDD more than its limit.	Update system regularly; replace HDD after 3 to 5 years of usage.

4. PC fails to detect HDD or BIOS	The power supply is giving out power that is above or below the required level, making it hard for the BIOS to detect the HDD so HDD fails to spin up.	Use UPS of a reputed brand. Replace IDE/SATA or SCSI cable. Switch the drive to another power plug.
5. Heat and dust effect	Excessive usage couple with poor ventilation. Faulty CPU fan	Ensure proper ventilation for the PC. Ensure the fan is working. Remove dust periodically by blowing out.
6. Manufacturing fault.	HDD works well initially, but then becomes non-responsive after a few months.	New HDD should be properly tested for functionality and reliability by using SMART tools that can notify user about the condition of the HDD and risk that the HDD may face in future.

HDD fault may also be signalled by an error message which may imply an issue with sectors data, or physical disk on the HDD. For instance in HP PCs with unified extensible firmware interface (UEFI), an error could appear as:

Hard disk error

Please run the hard disk test in System diagnostics

Hard Disk# (xxxx)

F2 - System Diagnostics

For more information, please visit:

<http://www.hp.com/go/techcenter/startup>.

In this case,

- ✓ Make an immediate backup of all important data and files
- ✓ Perform a HDD diagnostic test to verify the condition of the HDD driver, Run the HDD test in HP advanced system diagnostics using the steps in the section:

Although the majority of CD-ROM problems are caused by software or setup issues, the drives too are delicate and unforgiving devices. So it is better to replace a defective drive outright when it gets faulty rather than attempting a lengthy repair that their prices are fairly affordable. However, there are some situations that could require troubleshooting. Below are some symptoms and possible remedy in the course of operation of CD drive.

1.7.2 Troubleshooting CD-ROM Drives

1. Avoid bending the disc.
2. Protect the disc from heat.
3. Avoid scratch on the disc most especially circular scratches that follows the spiral tracks. A circular scratch can easily wipe out entire segment of data which would be unrecoverable.
4. Avoid use of chemicals on the disc; e.g. benzene, Ammonia acetone, carbon tetrachloride or chlorinated cleaners which can damage the disc's plastic surface.
5. Maintain comfortable environment; don't expose recordable discs to sunlight or other strong light for a long time. Also avoid high heat and humidity which can damage the physical disc.
6. Always keep blank or recorded media in a clean 'jewel' cases for best protection.
7. Don't write on the disc: ink might eat through the lacquer surface and damage data.
8. Don't use Kodak photo CDs: these CDs have a protection bit that prevents them from being written on many CD recorders. An attempt to write these discs on the recorders that recognize the protection bit will produce an error message.

Compact Disc records data in digital form through the use of physical 'pits' and 'land' in the disc. It is a remarkably long term storage media when handled effectively especially in storing it. Faulty CD can cause file and data errors which may appear as if it were the drive that is faulty. In order to maintain longevity of the disc, certain things need to be avoided. Things like:

1.7.1 Introduction

1.7 The causes of CD-ROM Drive failure

1. Turn the PC and immediately press the esc key repeatedly about once every second, to enter the start-up menu.
 2. Press the F2 key to open system diagnostics.
 3. On the HP Advanced System Diagnostics Screen, click Hard disk test.
 4. Click 'start hard disk test'.
 5. Wait for the test to complete, the result will be displayed on the screen.
- Contact customer support if necessary.

SYMPTOMS	POSSIBLE CAUSE	REMEDY
CD I drive can't accept or reject a A problem associated to motorized CD ROM drives.		➤ Check for any obvious obstruction through the CD slot. ➤ Check each linkage and motor drive gears carefully; replace any damaged gear or entire gear assembly or even the entire CD ROM drives mechanism.
2. optical read head does not seek	Could be power issue, or problem with the read head.	➤ Replace the CD ROM mechanism outright.

4. The Disc does not turn	
The disc is supposed to be turned at constant linear velocity (CLV) directed and regulated by the spindle under normal condition.	
> Check to be sure that the disc is seated properly and not jammed or obstructed > Before troubleshooting review your drive installation and setup carefully to ensure that the drive is properly configured for operation. > If symptoms persist, replace the CD ROM drive	> Check the CD itself and ensure that it is the right format, inserted properly and physically clean. > If the problem persists, check the physical interface cable between the drive and its adapter board to ensure that the cable is connected correctly and completely. > If you are using multiple SCSI devices from the same controller card, and other SCSI devices are operating properly, the SCSI controller is probably intact but if other SCSI devices also malfunction, then try a new SCSI host controller board; at this point either the drives are defective and your best course is to replace the drive.

	5. The optical head cannot focus its laser beam. The CD ROM drive must focus its laser beam to a microscopic precision to properly read the pits and lands of a disc. If the focus is out or not well maintained the laser detector might produce erroneous signals, randomly and consistent.	
mechanism.	➤ Check the disc to be sure it is optically clean from dust and fingerprints which could result into serious access problems. ➤ Try another disc if the same results occur, try cleaning the optical aperture.	

1.8 Cause of Display system Failure (CRT).

The cathode ray tube (CRT) type display system (monitor) is responsible for producing picture element and resolution. The CRT as extensively discussed in COM 216, has stationed electrodes and grids that processes electron beam into pictures or graphics that can be seen on the screen. The CRT receives four (4) inputs i.e; Video (RGB), Vertical sweep that scan the CRT from top to bottom, Horizontal sweep that scans from side to side and High voltage which attracts electrons to the screen to process image.

1.8.1 CRT Problems

The problems that could occur to the CRT may be due to physical shocks and/or aging of the display system. Physical shock could dislodge elements inside the CRT to create an issue. Also, regular and prolonged use of the display system could alter the physical dimension of the cathode and grids, resulting in the development of slower, more gradual short circuit.

Below are some faults that are associated with CRT display system and their remedies.

Nature of Fault	Symptoms	Kennedy
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1. Heater is open	Dead screen	* Replace the tube.
2. Heater shorts to cathode due to accumulations, oxidation or corrosion.	Image appear saturated with the color of the defective electron gun (resistance between heater and cathode should be infinite).	* Replace tube.
3. Cathode shorts to the control grid (due to corrosion flake-off of the cathode and contact to the control grid).	Control grid losses its effectiveness and the corresponding color will appear saturated similar to a heater short.	* Replace tube.
4. One or more colors appear weak (due to barium emitter developing layer of ions or wear out, inhibiting the release of electrons — 'cathode poisoning')	Affected cathode (s) losses efficiency resulting in weakened screen color.	* Replace tube.
5. Reduced sensitivity of images.	'Phosphor burn effect' dark shadow cast on screen after turning off the monitor.	* Keep brightness at minimum and use screen saver utility if an image will sit unchanged for a long time.
6. Magic appear to shake, or oscillates in		* Replace tube.

size.			
7. Images appear to flip or scroll vertically.	Vertical oscillator IC is faulty in the vertical circuit.	▪ Replace the IC. ▪ Check the power supply.	
8. Image expand or shrinks in horizontal direction when the monitor gets warm	Thermal related problem	▪ Replace IC ▪ Check power supply.	

Computer display monitors has evolved from the old cathode ray tube technology to LCD (liquid Crystal Display) Technology. An LCD screen is a type of flat-panel display technology that uses liquid crystals, sandwiched between two transparent electrodes, to control the amount of light that passes through them thereby creating images. The technology uses the Cold cathode Fluorescent Lamp (CCFL); a type of fluorescent lamp where light is produced by an electric discharge in mercury vapour, which then excites a fluorescent coating on the inside of the lamp thereby producing visible light without needing hot cathodes. There is also the LED type monitor. An LED monitor is a type of LCD display that uses Light Emitting Diodes for backlighting.

The most common problems associated with LCD monitors include:

- Broken screen
- White screen with no picture
- Screen has one or more colored vertical or horizontal lines
- Discoloration or "spots" on the display
- Flickering or Ghosting issues due to faulty capacitors, backlight, inverter, panel or board.

1.9 Causes of keyboard/ mouse errors

1.9.1 Keyboard

The keyboard is one of the most basic input device used to navigate through the execution of processes and programming with the help of keyboard device drivers which could be in-built into the OS or in the case of extended keyboard (s); installed separately.

Keyboards are of different make in terms of Technology. There are:

Scissors switch (for notebooks)

Full press membrane (for desktop)

Direct switch (for phones hand held devices)

Capacitance switch

Dome switch.

1.9.1.1 Keyboard failure symptoms

1. No character appears on screen as entered from the keyboard

2. Some keys work while some others don't.

3. Error messages appearing such as six (6) short beeps (AMI BIOS), 301 error code (for IBM compatible system).

4. Unplugged keyboard error appears

5. A keyboard is stuck.

1.9.1.2 Causes of keyboard error

a. Failure to re attach and re enable the keyboard from CMOS after disabling.

b. Wrong insertion of keyboard cable into the PC connector.

c. Positioning the keyboard switch in the "lock" position.

d. Problem with keyboard interface IC.

e. Broken connector at the power supply keyboard connector.

f. Broken connector's soldering junctions on the MOBO.

g. Electrical contacts coated with impurities thereby failing to make good contact.

1.9.1.3 Preventive measures

Keyboard problems do not happen suddenly unless the keyboard is physically abused. Dust and debris accumulation takes a long time to create serious repetitive keyboard problems. By adopting a regular cleaning habit most problems can be stopped before they even occur. It is advisable that in normal office environment keyboard should be cleaned once every 4 months, at

homes once every 2 months and more frequently in harsh industrial environments. Keyboards should always be covered with 'thin membrane skin' to prevent dusts.

1.9.2 Mouse

Mouse is a pointing device that provides easy means of selecting symbols (graphics, buttons or icons) and text on the VDU. It uses a combination of software and hardware to produce and control a graphical screen cursor.

The software device driver generates the cursor and reports its position on the screen, then interprets the motion of the hardware device corresponding to the movement of the cursor. Selection is done by positioning the cursor over a graphical symbol and activating a button on the mouse.

The mouse requires the following to work:

1. Physical signal generating hardware,
2. Software driver (device driver),
3. Application program written to make use of the device driver.

The basic function of mouse is to click, double-click, drag and scrolling. Varieties of mouse include:

Wireless or cordless mouse

G-stick mouse

Trackball mouse

Optical mouse

Mechanical; (or ball mouse). Etc.

1.9.2.1 Cleaning the mouse

The process of cleaning a ball mouse involves:

1. Remove the ball by rotating the retaining ring gently.
2. Clean the ball by washing thoroughly with soapy water, and then dry with towel.
3. Use photography-grade compressed air can to blow the dust out.
4. Clean the small pressure rollers and the X, Y rollers with a cotton swab dipped in isopropyl alcohol.
5. Re-assemble and test.

1.9.2.2 Causes of Mouse error, failure symptoms and remedy

Failure symptom	Possible cause	Remedy
1. Single mouse click works but double-click doesn't. 2. Attempt to double-click but gets quadruple-click, or attempt to single click and gets a double click. 3. Mouse stops working after a few minutes only to resume when the PC is reboot. 4. Mouse port disabled or mouse not present.	Double click speed is set too high. Likely a phenomenon called "button bounce" has occurred. This is as a result of a hardware defect due to broken or poorly buffered mouse buttons. Build up static charges (though not sufficient to damage the mouse) in the mouse that is interfering with the mouse circuitry causing it to stop responding. Mostly a connection problem or set up problem.	Lower the speeds {start>settings>open control panel>double click mouse icon and adjust the double-click speed slider under buttons tab} Clean the mouse buttons with good-quality electronic grade contact cleaner. Replace the mouse. ○ Spray the surrounding carpet and upholstery with a very dilute fabric softener to dissipate static built up. ○ Ensure that the PC and house wiring is grounded properly ○ Replace the mouse with a more static resistant model. Ensure the mouse cable is intact and is securely attached to the PS/2 port. Check the CMOS set up and see that any entries

for your mouse are enabled properly. Check the driver to be sure the appropriate driver for the mouse is in place. Otherwise change the mouse type under windows by removing the old mouse reference from the device manager, and then use the Add new hardware wizard to install the new mouse manually.	A wrong driver for the mouse has been installed on the PC.	5. Error message "this pointer device requires a newer version"
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2 COMPUTER SYSTEM PERIPHERALS

2.1 Peripheral Devices (PDe)

PDe, also known as peripheral, computer peripheral, input-output device or any of various devices (including sensors) used to enter information and instruction into a computer for storage or processing and to deliver the processed data to a human operator or, in some cases, a machine controlled by the computer. In other word, it refers to an input/output unit of a system that feeds data into and/or receives data from the central processing unit of a digital device.

Peripherals to a digital device include any device that is connected externally to the digital device, any device internal to the digital device that connects the digital device by wire or cable, and any circuit board designed for interchangeable mounting, internal or externally, that increases the operating or processing speed of a digital device; such devices makeup the peripheral equipment of modern digital computer system eg turbo cards and enhancement boards.

2.1.1 Types of peripheral devices

Peripheral devices are commonly divided into three (3) kinds:

1. Input devices

- ii. Output devices and
- iii. Storage devices.

An input device converts incoming data and instruction into a pattern of electrical signals into binary code that is understood by the computer. The output device reverses the process by translating the digitized signals into a form intelligible to the user.

Input devices include: typewriter like keyboard, hand held devices such as the mouse, trackball, joystick, track pad and special pen with pressure – sensitive pad, microphones, web cams, and digital cameras. They also include sensors that provide information about their environment- temperature, pressure etc.

Output equipment includes visual display terminals, inkjet and laser printers, loud speakers, headphones and devices such as flow valves that control machinery often in response to computer processing of sector input data.

Some devices such as video display terminals and USB hubs may provide both input and outputs functions. Other examples are devices that enable the transmission or reception of data between computers e.g. modems and network interfaces.

Most auxiliary storage devices such as CD-ROM drive, DVD drives, flash memory drives and external disk drives also double as input/output devices. Even devices like smart phones tablet computers and wearable devices like fitness trackers and smart watches can be considered as peripherals albeit ones that can function independently.

2.1.2 Standards for connecting peripherals to computers.

a. Serial advanced technology attachment (SATA): - This is the most common interface or bus or port that is used for connecting peripheral devices. A bus can be either serial or parallel. Serial bus sends bit at a time while parallel bus sends byte. Serial connectors have fewer wires and are generally simpler than parallel connections.

b. USB

2.2 Causes of Serial, Parallel and USB ports Failure.

As a general guide to resolving problems with devices that are connected to serial port or USB port, do the following:

- i. Verify that the port is enabled and is assigned a unique address
- ii. Verify that each bus is assigned a unique address
- iii. Verify that each port is assigned a unique address
- iv. Verify that the device is properly installed and firmly seated
- v. Verify that the serial port adapter, if you install one, is properly installed and firmly seated

- vi. If the failing device is a printer run the printer self-test
- vii. Run diagnostics on the port: A serial or parallel port device failure may require replacement of the system board, riser card or I/O board; if the component is found defective by diagnostics. For parallel port install a wrap plug in the port and run the diagnostic test to determine which field replacement unit (FRU) failed. If the diagnostic tests (with the wrap plug installed) cannot detect the failure, replace the cable. If that does not correct the problem replace the system board and adapter connected to the cable, or the component indicated as failing in diagnosis.

2.3 Causes of Ports failure and troubleshooting

Cause of failure	Kennedy
• Use of wrong cable. • Conflicts (caused by improper resource allocation in device manager). Verify that the cable being used meets the specification for the device being connected. Try using a different cable that is known to be good, to rule – out cable issues. Verify that ports have no conflicts, and are seen correctly in device manager. Conflicts exist if a yellow mark (!) is listed next to the item. Opening device manager requires the following steps: 1. Right – click on My computer. 2. Click properties. 3. Select hardware tab in the properties window. 4. Click the device manager button.	• Power the system, 2. Press F1 to enter IBM configuration set up, 3. Using the arrow keys select Devices and I/O ports then press enter. For serial port troubleshooting: • Ensure that the port is marked Enabled, and verify that the IRQ settings is correct for the port. • Set a unique IRQ for each SCSI, RAID or NIC

2.4 Cause of Printers' failure

The reasons for Printers failure include:

1. Poor power supply
2. Improper connection to the PC
3. Ink issues
4. Corrupted or bad printer Driver/software
5. Printing to the wrong printer

	install in the system if possible to prevent performance issues or hangs. ✚ Disable any unused devices to free up IRQ resources like serial ports, parallel port, on-board SCSI if not in use. When 'Disable' is selected, the system will not configure the disabled device and the OS will not see the device. ✚ Press escape to return to the main menu. - For the parallel port: ✚ Using the arrow keys, select parallel port set up the press Enter. ✚ Ensure the parallel port is set to Enabled (do not set it to PNP or Auto -- configure) ✚ Ensure the parallel port mode is set to Bi-directional (do not set it to HPP, HCP or EPP&HCP) ✚ Press ESC to return to the main menu. ✚ Select "save settings", then Exit the set-up utility. - Note: You can select "devices and I/O ports" choice to view or change the assignments for devices and input/output ports if need be. You can add serial port by installing a serial adapter in an expansion slot.
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6. Trying to print on a printer with "use printer offline" option enabled (windows only)
7. Paper jam due to wrong insertion of paper, paper quality, broken tray, damaged rollers (out - used) and leaking cartridges (low - quality cartridges tend to leak a lot, and the ink strays can cause paper jams after drying.
8. Clogged nozzles: If you are using an inkjet printer, the nozzles of the cartridge can get blocked, causing fading output "ghostly" prints.
9. When ink gets collected on one side of the tower, some parts of the page get printed perfectly while others are faded.
10. Fuser and toner problems (laser printers): laser fuser that warms up the toner. On heating, the toner sticks to the sheet and then creates a print. But the fuser has to be adequately attached to the toner for it to create ideal prints. If the toner sticks to the upper roller of the fuser, it will lead to ghosted prints. The reason being that when the toner is attached to the upper rollers of a fuser, whenever a print gets completed, some toner particles get left behind. So, when you give any subsequent print command, the toner particles from the previous print get deposited over the new one. This appears as a strange image called a "ghost image". Separating the toner from its upper rollers will solve the problem.
11. Low ink density: in many cases where the printer starts giving faded outputs, the ink/toner density ratio is set to very low values. All you need to do is go to "Devices and printers" settings and increase the ink/toner density slightly. In some printers the default ink /toner density is low. Therefore, as soon as you get the printer, you should increase the density slightly to prevent fading problems at least for a few years.
12. Contamination of cartridges leading to ink smearing: contaminants like dust, hair etc may find their way into the ink cartridges, so when they accumulate in large quantities, they get dragged across the sheet during every print. If the cartridge were not cleaned, the instances of misprints would keep increasing.

2.5 Printer self-test

Printer status can be known; especially portable printers by doing the following:

Turn the printer power on by simultaneously holding down both the feed and power button for 5 seconds or more (until printing begins). The printer will print the current printer status, including the firmware version, communication mode and print setting.

3.1 Definition of Virus

A true virus is a piece of code that attaches itself to an executable file and is not activated until the executable file is launched.

3.1.1 Examples of Virus

Viruses can be categorized by where they hide themselves in the computer system. The types are:

1. File virus: - This is the most common type of virus. This virus hides itself in executable file such that when the executable file is run, then the virus is activated. Example include.....
2. Macro virus: - These viruses attach themselves to portions of application and disguise themselves as macros (an automated process within an application such as reading and updating a data field or searching for and formatting specified text). Example include ...
3. Boot sector virus : - This type of virus hides itself in the Master Boot record (MBR) and is activated during start up when the MBR is located and initialized. Example include.....

3 VIRUS PROTECTION UTILITY FAILURE AND REMEDY

3.1 Virus protection utilities failure.

Virus(s) normally replicates themselves into memory when they are introduced into computer system either through floppy disks or flash disks, downloading files from the internet or executing email attachments that launch a host program such as Word processing.

Antivirus programs are mainly designed to prevent viruses from having their way into the computer system. Unfortunately, no system is completely immune to computer viruses because new viruses are created all the time and could beat the ability of the antivirus utility in present use to detect it, and also due to the fact that in most cases, antivirus utilities work by recognizing and removing specific viruses; they are not useful against viruses that have been created since the release of the utility itself.

When virus protection utility fails, the following are likely to happen as an indication:

- i. Display of particular meaningless messages; like bouncing ping pong balls on the screen.
 - ii. Running a video clip intermittently
 - iii. Changing of display colours
- The above may be considered as non - destructive payload. More challenging effects of viruses which are considered as destructive payload include:

- a. Deleting of files
- b. Closing of running application
- c. Destruction of MBR
- d. Unexpected crashes, etc.

3.3 Remedies

1. If antivirus utility is present, run it and instruct it to perform a virus scan and removal.

2. Keep up with updated versions of antivirus utility software in order to greatly reduce the risk of being infected with newer trend of viruses.
3. If the antivirus in the system could not detect and remove the virus, and it appears the virus is limited to the boot sector, use FDISK/MBR command. This could possibly replace the infected MBR with a good copy from the floppy disk.

4 NETWORKS AND NETWORK SET UP

4.1 Computer network

A computer network is a set of computers sharing resources located on or provided by network nodes. It refers to interconnected computing devices that can exchange data and share resources with each other. These network devices use a system of rules called communication protocols to transmit information over physical or wireless technologies. See figure 4.1 and 4.2.

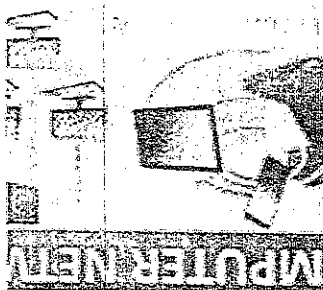


Figure 4.2: Wireless computer network

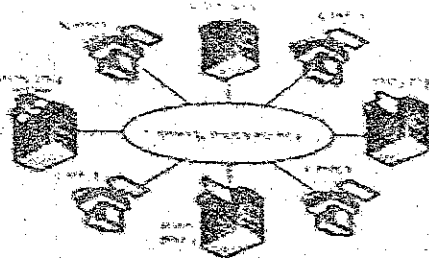


Figure 4.1: Wired computer network

Usually, networks are categorized into wired and wireless. The wired network uses physical connection with cables that limits its application to personal area network (PAN) or local area network (LAN) in most cases due to distance barrier. On the other hand, the wireless network can be in form of wireless LAN, wireless metropolitan area network (WMAN), wireless PANs, wireless wide area network and campus area network.

4.2 Hardware and Software components of a computer network

The major hardware components that are used in a computer network are: Server, client and peer, transmission medium and connecting devices. The software components are basically, the operating system and protocols.

4.3 Advantages of computer network

- Resource sharing; which is the main purpose of a network.
- Cost reduction: sharing devices such as printers saves money; more so, site (software) licences are likely to be cheaper than buying several stand-alone licences.
- Network users can communicate by email and instant messengers like whatsapp, Tiktok etc.
- High reliability
- Improved performance; files can easily be shared between users .etc.

4.4 Network Topologies

A network topology describes how the nodes or devices (physical or virtual) in a network are arranged; and how they communicate with each other.

The following are the common network topologies:

- a. Bus topology: - This is used in small networks. Every node is connected along a linear path.
- b. Ring topology:- Nodes are connected in a loop, and traffic may flow in one direction or both directions.
- c. Star topology: - A central node connects to all other nodes in the network. It is a common and stable topology used in LANs.
- d. Mesh topology: - The nodes are linked in such a way that multiple paths between nodes are possible. It increases the resilience of the network but also increases cost. A network could be fully meshed (all nodes connecting to all other nodes) or partially meshed (only some nodes having multiple connections to other nodes).
- e. Spine – leaf (tree) topology: - Multiple star topologies are connected together in a larger star configuration.
- f. Hybrid topology: - A combination of other topologies used within one network.

4.5 Network set up (configuration)

A network configuration is the process of assigning network settings, policies, flows and controls. In a virtual network it is easier to make network configuration changes because physical network devices appliances are replaced by software; removing the need for extensive manual configuration. Network configuration can also be automated and managed via a centralized configuration manager called network configuration manager. This manager reduces IT work load and makes it easier to:

1 maintain network

2 make configuration changes

3 re-launch devices, and

4 tracks and report data.

Some basic network configurations include:

- Switch/router configuration
- Host configuration
- Software and firewall configuration, and
- Network topology.

4.5.1 Importance of network configuration

The right configuration is essential to supporting the flow of traffic through a network, and can also support and enhance network security, network stability. The use of network configuration manager gives the following benefits:

1. Automatic data tracking and reporting allows administrators to spot any configuration changes and potential threats or issues.
2. The means to quickly roll back network settings to a previous configuration.
3. An easy way to make bulk changes such as blanket password change in a situation where passwords are compromised.
4. Reduce down time due to increased visibility and ability to quickly identify changes.
5. Streamline maintenance and repair of network devices (physical or virtual) and connections.

4.5.2 Network setting procedure

- > Select the start button, and then type settings. Select settings > Network & internet. The status of your network connection will appear at the top.
- > For windows 10, select start > settings > network & internet > status > network and sharing centre.
- > In windows 11, select start > type control panel, then select control panel > network and internet > network and sharing centre.

4.5.3 Setting up a wireless connection (WI - FI) in Windows 7

1. Click the start button and then click control panel.
2. In the control panel window click network and internet
3. In the network and internet window click network and sharing centre
4. Under change your network settings, click set up a new connection or network

5. In the window, click to select "manually connect to wireless network" and then click the "Next" button.

6. In the window enter the appropriate Network name, security type, encryption type and security key. Note however that the network name (SSID), security type, encryption type and security key information are configured on the wireless router or access point. If you are using a default configuration, this information may be listed on bottom of the router or in the manual. If you configured the router with custom settings, you may be able to locate it by logging into the router set up page. If you can't find this information, contact your network administrator or the router manufacturer.

7. Verify that the box is checked next to "start this connection automatically". If you are manually configuring your wireless network because the router isn't broadcasting the SSID. You must check the box next to "connect even if the network is not broadcasting".

8. In the successfully added window click on close button. If after all these steps you still can't connect, and then verify the network name (SSID), security type, encryption type and security key are correct and repeat the procedure.

4.5.4 Checking Network Configuration

In a command prompt environment, use ipConfig (for Windows), or ifConfig (for Linux network configuration). With a network configuration manager or with APIs you can check and set up the network configuration in a centralized software interface, allowing you to more readily configure, monitor and administer your network.

4.5.5 Configuration of network switch and Router

Some of the configurable setting on a network switch and router include:

- IP address : - for identification
- Password : - for added security
- Channel and band selection : - to improve performance.
- Default gateway : - to make the device visible to network management tools.
- Neighbour discovery : - for added visibility
- Correct time : - for proper troubleshooting and detailed error logs.

Note that a network configuration manager is the easiest way to perform network switch configuration and apply these settings consistently to every device on your enterprise network.

4.6 Network Operating Systems

A network OS is a computer OS that is designed primarily to support work stations, PCs and in some instances older terminals that are connected on a LAN. The role of a Network OS is to

provide basic network services and features that support multiple input requests simultaneously in a multiuser environment. The software behind a network OS enables multiple devices within a network to communicate and share resources with each other.

There used to be two basic types of network-OS. They are: Peer -- to -- peer (P2P) OSs, and Client server network OSs.

The P2P type allows users to share network resources saved in a common accessible location. All devices are treated equally in terms of function ability. It is best for small and medium LAN. The client server network OSs provide user with access to resources through a server. The network is controlled centrally which makes changes or addition to technology easier to incorporate.

However, a typical network OS no longer exist currently. The reason is, most OSs have built -- in network stacks that support a client-server model.

How to Fix a Modem That Won't Connect to the Internet

1. Check the lights on your modem. ...
2. Check the power supply and coaxial cables. ...
3. Reboot your router and modem. ...
4. Check your Ethernet cables. ...
5. Disconnect all connected devices. ...
6. Plug your computer into your modem. ...
7. Keep your modem cool. ...
8. Update your modem and router's firmware.